Sourcing list

DIY CNC epoxy-granite mill · October 2026

Parts to buy for the machine in `model/DIYCNC.FCStd` / `docs/5-design/report.txt`, sourced on Amazon (US) where it makes sense. Prices and stock were checked on **2026-10-09/10**. "?" means the price was not visible (Amazon blocks automated page reads); check the live listing. Shipping and tax are not included. Off-Amazon picks are marked "not Amazon": they are there because Amazon doesn't carry the part or is much dearer. Rows marked (opt) or (choice) are left out of the totals, and so is the "add later" section of Part 1.

The same list is in `docs/1-sourcing/sourcing.csv` for a spreadsheet.

Totals

The list is in three parts, so you can buy and build in that order: **Part 1** is the machine itself without the expensive equipment (castings, steel, rails, ballscrews), **Part 2** is the motors, controller, wiring and spindle, and **Part 3** is the enclosure and flood coolant. The machine runs after Part 2 with brush-on cutting fluid.

Add later (niceties). The way covers, the screw cover, the drip shield, the stainless sump and the grease gun bolt on whenever you buy them: everything they fix to is already cast in (inserts and a tapped pad for the roller feet, inserts for the cable-chain mount, the beam studs for the drip shield) or needs no fixing, and so are the inserts, gutters, drain and sump space for Part 3. Fit the way covers before steady steel work. Anything cast in (the alumina grit, the inserts, the drain) has to be decided before the pour.

Part	Section	Priced
1	Castings and hardware	$1,309.11
1	Rails, screws and mechanics	$247.14
1	First chips	$20.00
1	Add later (niceties), not in the frame total	$183.09
2	Motors and drivers	$232.12
2	Controller and wiring	$186.19
3	Enclosure and flood coolant	$499.23

Part	About
1. Frame	**$1,576.25** + $183.09 of niceties, added later

Part	About
2. Electronics + spindle	**$418.31** + spindle: $250-375 (1.5 kW 110 V kit) or $430-600 (2.2 kW + 120 V step-up VFD)
3. Enclosure + flood coolant	**$499.23** + 1 unpriced (Flood coolant pump): about $650-700
Total	**about $2.89-3.29k**, plus the niceties

Shipping and tax are extra (Raka ships free over $100). The whiting and Bolt Depot ship heavy or separately; buying the whiting and the steel locally saves most of it. Steel-yard and remnant prices are estimates: call a yard with the cut list.

JLCMC and JLCCNC (checked 2026-10-10). JLCMC list prices are about half of Amazon for rails and couplings (e.g. the Y rail pair $27.40), but US orders prepay a 55 % duty plus DHL shipping, so the whole motion order comes to about $250-300 against about $190 here. There is no ready-made SFU1204 (lead 4) kit, and the 1605 screw-end machining for BK12/BF12 is not confirmed. Worth adding the rails and couplers only to a JLC order you are placing anyway. JLCCNC machining of the three aluminum plates with all holes is estimated at $430-850 delivered (90 % duty prepaid on machined aluminum): not worth it for plates drilled to the parts (and the X carriage and Z plate are steel now). Retail cut-to-size shops (e.g. Metal Supermarkets, about $820 for the tube, risers and aluminum) also cost more than the sources here.

Part 1 - Frame: castings and hardware

Item	Spec	Qty	Listing	Price	Line	Notes
Epoxy	~10.5 kg low-viscosity laminating epoxy (8.5 % mix), plus ~10 % waste	1 (3 gal)	Raka Standard 2:1 Epoxy Laminating Kit: 127 Thin resin + 606 Slow hardener, 3 gal (not Amazon)	$172.99	$172.99	About 12.5 kg, $13.8/kg; free shipping (Raka: orders over $100). Cast the test bars first; if the mix needs more than 8.5 %, add the 1.5 qt kit ($32.99). Alt: US Composites 635, 4 gal $245. Avoid deep-pour casting resins and 1:1 tabletop epoxies.
Fines	Whiting (calcium carbonate) 325 mesh, ~11 kg	1 (50 lb)	Clayworks Supplies calcium carbonate whiting, 50 lb, pickup (4625 Falls Rd, Baltimore; not Amazon)	$17.00	$17.00	Call first (410-235-5998) to check it is 325 mesh. Shipped alt: Clay-King, $0.84/lb plus heavy shipping. No crystalline silica (check the SDS), so a basic N95 is enough. Avoid garden lime and talc.
Aggregate	~107 kg: fine sand 17, coarse sand 32, pea gravel 23, 3/8 in gravel 24 kg, plus test bars; and 11 kg more pool filter sand, dry, to fill the tubes	~8 bags	Pool filter sand + Quikrete all-purpose sand + Vigoro pea gravel + 3/8 in stone (Home Depot)	~$9/bag	$71.00	Buy locally. Wash and dry fully. The same plain mix in every casting.

Item	Spec	Qty	Listing	Price	Line	Notes
(opt) Alumina grit, coarse	Brown fused aluminum oxide #16 (12-20 is fine), ~19 kg (18.1 net): the coarse sand in the beam and uprights	1 (50 lb)	IMS Blasting brown aluminum oxide, 50 lb (pick 16 grit)	$109.99	$109.99	Not recommended: it adds about 8 % to the frame, but the frame is only 2-5 % of the deflection at the tool, so under 0.5 % there. Decided before the pour if at all (it is cast in). Free shipping. Alt: NT Ruddock or Precision Finishing, about $80 + freight; Kleen Blast AO5016 $90 + shipping.
(opt) Alumina grit, fine	Black aluminum oxide 70 grit, ~10 kg (9.3 net): the fine sand in the beam and uprights	1 (25 lb)	Central Pneumatic black aluminum oxide 70 grit, 25 lb (Harbor Freight, item 56704)	$39.99	$39.99	Upgrade on top of the #16, decided before the pour: about +5 % more. Without it, the fine sand stays sand (in the aggregate row).
Mold board	3/4 in white melamine (a true 0.750 in), about 2.2 m2 of boards: cut list and layout in the mold guide	1 (4 x 8 ft)	Veranda melamine white particleboard panel, 3/4 in x 4 ft x 8 ft (Home Depot, model 461877)	$43.98	$43.98	One sheet: one upright mold casts both uprights, and the bridges and struts come out of the offcuts (three-stage cut in the mold guide). Have the store rip the long strips. Measure the thickness: the printed saddles and brackets are drawn for 19.05 mm (mold_board in the model).
Mold cleats	1x2 lumber, 5.0 m: the strips along the walls' feet	3 (8 ft)	1x2 furring strip (Home Depot / Lowe's)	~$2.50	$7.50	They lie on the floor outside the walls and never touch the mix.
Mold hardware	M4 x 30 (87), M4 x 35 (30), M4 x 45 (33), M4 x 20 (4, leg pinch screws), M4 nuts (154) and washers (150); M3 x 25 with nuts and washers (20)	packs of 100 / 50	Zinc-plated metric (Zoro, Bolt Depot or a hardware store; Zoro M4 x 30 is $9.89 per 100)	~$40	$40.00	Exact counts in the mold guide; add about 10 % spare. Reused for every casting.
Mold rods and bolts	Plain steel rod: 1/2 in 2x 3 ft; 1/2-13 x 1-1/2 in bolts (2), 3/8-16 x 3 in (2); M5 x 20 (17, insert screws), M5 x 30 (12), M5 x 50 (8), M6 x 30 (6), M6 x 50 (6)	1 set	Everbilt plain steel rod and hex bolts (Home Depot); the metric screws in packs online	~$40	$40.00	Greased, they form the beam's stud holes and the dowel holes and hold the steel and the inserts to the mold boards. Reused.
Mold release wax	Partall Paste #2	1	Partall Paste #2, 12 oz (Fiberglass Supply)	$20.96	$20.96	3-5 coats, then the PVA.
Mold release film	PVA, over the wax	1 (pint)	Partall #10 PVA film (Fiberglass Supply)	$8.36	$8.36	
Edge rounds and fillets	3/8 in PVC quarter-round, about 6 m; a tube of silicone	1 set	PVC quarter-round molding + clear silicone (hardware store)	~$20	$20.00	Quarter-round in the corners that make outside edges; silicone fillets elsewhere (the mold guide).

Item	Spec	Qty	Listing	Price	Line	Notes
Steel, stock flat bar and tube	Flat bar 3/8 in x 1-1/2 (4.2 ft), x 2 (2.8 ft), x 2-1/2 (8.4 ft), x 3-1/2 (one 3-9/16 in piece, the Y bracket pad); 1/2 x 2-1/2 in (2.6 ft); 1/2 x 5 in and 3/8 x 5 in (20 in each, the beam); 5/8 x 6 in (16-1/16 in, the X carriage, and 11-13/16 in, the Z plate). Angle 1-1/4 x 1-1/4 x 1/4 in (2x 10-1/8 in, up the back of the carriage). Tube: 2x3x1/8 (4.0 ft), 2x2x1/8 (1.8 ft + 4x 8-1/8 in), 1-1/2x1-1/2x1/8 (3.2 ft + 4x 6 in)	~151 lb	A local steel yard or Metal Supermarkets, cut to the list in docs/1-sourcing/steel/ (Online Metals, Speedy Metals and Zoro sell the same sizes by the piece)	~$375	$375.00	About $2/lb for flat bar and $3/lb for tube, plus cutting, at a yard or service center: get a quote with the cut list (Baltimore: Alro Metals Outlet, Essex; Metal Supermarkets, Golden Ring Rd; DS Pipe & Steel, Wicomico St). The beam bars are 1/2 and 3/8 in and the 2x2 tube 1/8 in wall: thicker bought under 0.5 % at the tool (design notes). The X carriage is steel too (about 20 lb with its angles, $40): it was the softest part between the tool and the work (design notes). So is the Z plate (about 13 lb, $25): stiffer and cheaper than the aluminum bar. Speedy Metals: 3/8 x 2-1/2 in flat $19.13 per 36 in (about $2/lb), plus shipping, and you cut them. Retail cut pieces cost far more (Zoro: about $27/ft for 3/8 x 2-1/2 in bar). Drill and tap from docs/1-sourcing/steel/steel-templates.pdf.
Grinding	Blanchard grind both faces flat and parallel: the X carriage, the Z plate and the two 5/8 x 1-1/4 x 300 mm Z risers	1 lot	Metal Supermarkets (grinds what it sells at some stores; ask with the steel quote) or a local grinding shop	~$75	$75.00	Before drilling. The rails and blocks sit on these metal to metal, with no epoxy bed to take up mill scale or bow; the Y and X risers are bedded on epoxy, so they need nothing. Buy the Z risers with the plates so one lot covers all four.
Rail risers	Cold-rolled 1018 x 1-1/4 in: 3/4 thick 2x 600 mm; 5/8 thick 2x 400 + 2x 300 mm	~4 + 5 ft	Steel yard (local)	~$40	$40.00	About 26 lb at $1-2/lb. Online: Speedy Metals $60 + shipping. Pieced bars are fine (two rail screws per piece).
Aluminum table	6061 flat bar in a stock width, cut to length: 3/4 x 10 x 12-5/8 in	~9 lb	Drops or flat bar from the steel yard / eBay "6061 flat bar"; Speedy Metals cuts the same bar to length	~$45	$45.00	A stock width, so a drop or a piece of bar only needs a length cut. About $2.3-4.5/lb if you find drops. Bought new and cut to length it is about $105 + shipping (Speedy Metals by the inch, 2026-10-26: 3/4 x 10 in $8.31/in). Face it on the machine once it runs. The X carriage and Z plate are steel now (steel row).
Threaded studs	1/2-13 grade B7 fully threaded studs, bought at length: 4x 14-1/4, 2x 6 in (uprights), 2x 27 in (base draw rods), 4x 10-3/4 + 6x 8-1/2 in (base cross rods)	18	Bolt Depot B7 fully threaded studs, plain (sold in 1/4 in steps; not Amazon)	~$2-10	$80.00	Nothing to saw: every rod is a stock stud length (each within 3 mm of the length it replaces; the tail past the far nut takes it). Estimate from the 6 in at $2.18; longer ones scale with length. B7 is about as strong as grade 8, well above the zinc rod it replaces ($60 at Home Depot, sawn by hand). If a length is not stocked (the 27 in draw rods most likely), cut it from 1/2-13 B7 rod.
Beam studs	3/8-16 grade B7 fully threaded studs, bought at length: 8x 7-1/2 in (beam studs), 3x 4-1/2 in (beam ties); 3/8-16 nuts and washers (11)	11	Bolt Depot B7 fully threaded studs, plain, 3/8-16 (not Amazon)	~$1.5-2	$20.00	Same order as the 1/2-13 studs. Grade B7 is about as strong as a grade 8 bolt; plain zinc rod is not. If not stocked at these lengths: 3/8-16 B7 rod, 3x 3 ft ($7.63 each), sawn.
Pipe nipples	1/2 in x 2-1/2 in black steel pipe nipple (beam spacers)	3	Everbilt black pipe nipple (Home Depot)	~$3	$9.00	
Bolts for the base grid	1/4-20 hex bolts with nuts and washers: 24x 4 in (rail bars onto the girders), 4x 2-1/4 in (pads)	1 set	Hardware store	~$15	$15.00	

Item	Spec	Qty	Listing	Price	Line	Notes
Nuts and washers	1/2-13 nuts x32, 1/2 in SAE washers x32, 3/8 grade 8 washers x8	5 packs	Everbilt packs (Home Depot)	~$28	$28.00	Nuts 15-pack $4.89; 1/2 washers 25-pack $9.75; 3/8 washers 25-pack $3.84
Dowel pins	3/8 x 1-1/4 in (6) and 3/8 x 1 in (4, base split)	2 (10-packs)	Steel dowel pins (Zoro)	~$12	$11.58	They only locate the castings
Feet	1/2-13 x 2 in hex bolt + jam nut + rubber pad, per leg	4	Hardware store bolts + rubber anti-vibration pads	~$12	$12.00	The bolt screws into the nut in each leg end; turn it to level, lock with the jam nut. Leveling mounts (S&W NYL B24, $9.49 each) if you prefer.
Rod sleeves	Clear vinyl tube 1/2 in ID x 5/8 in OD, 2.5 m: over the draw and tie rods inside the sand-filled tubes	1 (10 ft)	Clear vinyl tubing 1/2 in ID x 5/8 in OD, by the foot (hardware store)	~$6	$6.00	Cut to each tube's inside length. The rods come out later and the sand stays put.
O-ring cord	NBR 3 mm, 2.5 m	1 (3 m)	uxcell nitrile round strip 3 mm x 3 m	$6.56	$6.56	
Cord splice glue	Cyanoacrylate	1	Loctite 406, 20 g	$9.86	$9.86	
M5 cast-in inserts	19: the enclosure (11), the way-cover rollers (4), the beam cable-chain mount (2) and the Y cable chain (2). Nothing structural threads into the epoxy granite	1 (30)	uxcell knurled brass M5 x 16 mm	$11.08	$11.08	Screw each to the mold floor (wall) with a greased screw so it ends up flush. Cast them in even if the covers come later.
Low-head screws	DIN 7984 M5 x 20 (8, X risers) and M5 x 25 (10, Y risers)	18	DIN 7984 low-head, class 8.8, by the piece (Olander, Bolt Depot or McMaster-Carr)	~$15	$15.00	Head 3.5 mm tall, sits below the rail in the 4.5 mm counterbore. Zoro sells them only in 100-packs ($9.99 + $14.29).
Support screws	M6 x 40 (12: the Y and X BK/BF into their tapped steel)	1 (100)	Zoro M6x40 100-pack	$10.15	$10.15	The casting is relieved behind each tapped hole, so they never bottom
Angle screws	M6 x 16 button head (10): the carriage angles into the steel carriage	10	Hardware store	~$4	$4.00	Drill the angles, clamp them on and spot the carriage through them, then tap it M6.
Rail screws	M4 x 16	1 (100)	iexcell M4x16 12.9	$8.26	$8.26	
Block screws	M4 x 20	1 (50)	iexcell M4x20 12.9	$7.86	$7.86	Check length against the plate thickness
PETG filament	1.8 kg (sliced; 1.1 kg with the base's bevel strips from a 2x3): mold kit, core posts, formers and fillers, cast-in parts and machine parts	2 (1 kg)	SUNLU PETG 1 kg	$15.99	$31.98	The pocket cores are foam (next row). Printing them instead (foam=False in the model) needs about 1.4 kg more.

Item	Spec	Qty	Listing	Price	Line	Notes
XPS foam board	2 in x 2 ft x 8 ft: the base pocket cores	1	Owens Corning FOAMULAR 250, 2 in x 2 ft x 8 ft (Home Depot, model 24DD)	~$25	$25.00	Plus packing tape and double-sided carpet tape (~$10). Sizes in the mold guide.
Foam tape	Double-sided carpet tape + packing tape	1 each	Hardware store	~$10	$10.00	
Drain nipple	1/2 NPT brass	1 (2-pack)	Legines brass barb 1/2 NPT x 5/8	$10.99	$10.99	Cast into the base, pointing down into the sump. +$7.59 shipping; a hardware-store 1/2 NPT brass nipple works too.

Priced subtotal: $1,309.11

Part 1 - Frame: rails, screws and mechanics

Item	Spec	Qty	Listing	Price	Line	Notes
Y rails + blocks	HGR15 x2 at 600 mm + 4x HGH15CA	1 kit	iMetrx HGR15 kit, 2x 600 mm + 4x HGH15CA	$55.99	$55.99	Used at full length, no cutting. The listing calls the blocks "aluminum", which is probably wrong: check them for play. JLCMC alt: 2x E-BHH15C-2-L600-ZA-C-E30 $27.40 + 55 % US duty prepaid + shipping.
Z rails + blocks	HGR15 x2 at 300 mm + 4x HGH15CA	1 kit	BEVDICNC 2x HGR15-300 + 4x HGH15CA	$35.69	$35.69	Alt: CNCMANS B09N73DY32 $34.98
Y ballscrew kit	SFU1605-550, BK12/BF12, DSG16H	1	Ballscrew kit DSG16H + SFU1605-550 + BK/BF12	$39.39	$39.39	Ships with a 6.35x10 coupler: use the 8x10 below
Z ballscrew kit	SFU1204-300, BK10/BF10, DSG12H	1	CNCYEAH SFU1204 300 mm kit	$32.69	$32.69	Ships with a 6.35x8 coupler: use the 8x8 below. Its BK10 is a spare (the HM10-57 bracket is the bearing seat)
Z motor bracket	NEMA 23 + BK10 (HM10-57)	1	ANWOK NEMA 23 steel bracket, SFU1204	$32.02	$32.02	Same bracket as your X one
Y motor bracket	NEMA 23 + BK12 (HM12-57)	1	ANWOK NEMA 23 steel bracket, SFU1605	$31.36	$31.36	The bracket is the bearing seat: the kit's BK12 is a spare, and the Y pad is drilled for the bracket's six holes. A version with angular-contact bearings (C5) is stiffer along the screw. Alt: CNCMANS HM12-57 B09QS2S3BT $28.98. AliExpress HM10-57 / HM12-57 brackets are about $10-15 each (2-4 weeks): about $35 less for the pair.
Couplers	8x10 mm (Y) and 8x8 mm (Z), D25, two of each; the HM brackets take L30-L42 (the model fits a 42 on Y)	4	AliExpress "BW three jaws D25L30" plum coupling (not Amazon, 2-4 weeks)	~$1-3	$8.00	The spares cover X if your kit came with 6.35x8. They aren't needed until the axes go on, so order with the first parts. Check that D25 fits inside the bracket. Amazon alt: CNCTOPBAOS 8x10 3-pack B082TT7B84 $13.99 and Yeebyee 8x8 2-pack B0GTV7VMZB $12.89.
Grease and syringe	Lithium NLGI 2 + a grease syringe with a needle tip, for the blocks' M4 nipples	1 each	Lithium grease tube + needle-tip grease syringe (hardware store or Amazon)	~$12	$12.00	Enough for the blocks until the grease gun (add later).

Item	Spec	Qty	Listing	Price	Line	Notes
(opt) M4 grease nipples	M4x0.7 straight	1 pack	HARFINGTON 15 pcs M4x0.7	$13.94	$13.94	Only if the block nipples are missing or the coupler won't seat

Priced subtotal: $247.14

Part 1 - Frame: first chips

Item	Spec	Qty	Listing	Price	Line	Notes
Cutting fluid	Brush-on / drip, until flood coolant (part 3)	1	Tap Magic or similar cutting fluid	~$12	$12.00	Brush or drip it on; the base gutters and drain carry it down.
Drain bin	Any plastic bin or tub that fits under the drain, between the legs	1	Hardware store	~$8	$8.00	Until the stainless sump (add later) or the flood coolant (part 3).

Priced subtotal: $20.00

Part 1 - Frame: add later (niceties)

Item	Spec	Qty	Listing	Price	Line	Notes
Way cover strip	0.1 mm 301 stainless	1	SPCCGG 301 strip 0.1 x 300 x 1000 mm	$54.10	$54.10	Gives all three strips. If it turns out soft, use 304 full-hard B09ZQQJ34Z for the 85 mm strips. Fit the covers before steady steel work: chips wear the Y screw and rails.
Way cover springs	Constant force, about 0.5 x 30 in, 2.6 lb, for the printed rollers	4	AndyMark am-4666 constant-force spring	$6.50	$26.00	One per narrow strip, two for the wide one; plus shipping. Alt: SUS301 5-pack (Amazon B0DMM8YSMM) $35.99, or tape-measure springs.
Spiral screw cover	20 mm ID, size 20-250-40-28	1	Spiral steel bellows cover (eBay, US seller)	$21.85 + $4.99	$26.84	Clamps between the nut housing and the Y bracket's bearing end: no fixing in the casting. Amazon: MISONONEIK B0GTDV8DZN about $47.
Y cable chain	Printed PETG drag chain, 38 x 21 outer (as on X), about 20 links, standing on edge in the gap between the table and the right upright	1	Print it (print/ has the X chain links)	$0	$0.00	For a 4th or 5th axis on the table. Its fixed end screws to the two M5 inserts cast into the base top, its moving end into the two M5 holes tapped in the table's right edge.
Roller screws	M5 x 12 button head (6): the roller feet into the cast-in inserts and the tapped BK pad	6	Hardware store	~$3	$3.00	The fixings are already in the base.

Item	Spec	Qty	Listing	Price	Line	Notes
Drip shield	518 x 79 mm, 3 mm ACM	0	Cut from the ACM sheet offcut (part 3)	$0	$0.00	Clamped under the front beam stud nuts: no other fixing.
Sump	Full-size steam-table pan, 4 in deep (~13 L)	1	Winco SPJH-104, 22 ga stainless (not Amazon)	$22.16-26.81	$26.81	Not found on Amazon; restaurant-supply stores (KaTom, GoFoodservice $23.63) or a local one. Slides in under the base, between the legs.
Chip basket	1/3-size pan, 2-1/2 in deep	1	CAC STPT-S25-2 1/3 pan (not Amazon)	$9.98	$9.98	Solid pan: drill 6 mm holes in the floor (or buy a perforated 1/3 x 2-1/2 in pan). Sits at the front, under the drain.
Grease gun	Mini, 3 oz cartridges	1	Performance Tool W54205 mini grease gun kit	$17.59	$17.59	
Needle tip	For the blocks' M4 nipples	1	LubriMatic 05-037 needle	$6.98	$6.98	
Grease cartridges	Lithium NLGI 2	1 (3x3 oz)	LubriMatic 11312, 3x 3 oz	$11.79	$11.79	

Priced subtotal: $183.09

Part 2 - Electronics: motors and drivers

Item	Spec	Qty	Listing	Price	Line	Notes
X/Y closed-loop kits	23HS30-5004D-E1000 2.0 Nm + CL57T	2	STEPPERONLINE closed-loop kit 2.0 Nm + CL57T	$81.00	$162.00	Exact model. Low stock when checked.
Z driver (owned motor)	DM542T	1	STEPPERONLINE DM542T	$32.99	$32.99	Check that your motor is 4.2 A or less
Motor power supply	48 V, 3 drivers	1	Mean Well LRS-350-48 (7.3 A)	$37.13	$37.13	Trim to about 46 V: CL57T max input is 48 V

Priced subtotal: $232.12

Part 2 - Electronics: controller and wiring

Item	Spec	Qty	Listing	Price	Line	Notes
Controller	FluidNC, 6 axes, external drivers, RS485, 0-10 V	1	Elecrow Corgi ESP32 FluidNC controller (not Amazon)	$49.00	$49.00	Runs gSender or the web UI. Drives an RS485 VFD directly, so no 0-10 V module is needed for one.

Item	Spec	Qty	Listing	Price	Line	Notes
Limit switches	Roller lever, 4 needed	1 (5-pack)	Yusheng ME-8108 limit switch set of 5	$16.99	$16.99	Inductive LJ12A3-4-Z/BX (about $4.50 each) also work; power them from 12 V.
E-stop	22 mm mushroom, latching, 1 NO + 1 NC, in a box	1	BAOMAIN 22 mm E-stop station, keyed, IP65	$9.99	$9.99	Wire it to cut drive enable and VFD run in hardware. Was a $22.39 eBay station; the same switch.
Mains inlet	IEC C14 with switch + fuse	1	IEC C14 inlet module (Walmart)	$7.99	$7.99	Often ships without the fuse: buy 5x20 mm fuses.
Logic power supply	24 V, 6.5 A	1	Mean Well LRS-150-24 (TRC Electronics)	$20.39	$20.39	Jameco about $18.80
Electrical box	ABS IP65, 11.8 x 9.8 x 4.7 in	1	LeMotech ABS IP65 junction box 11.8x9.8x4.7	$25.99	$25.99	Holds the controller, three drivers and both supplies on a plywood or aluminum backplate; the VFD mounts outside. Steel alt: VEVOR 12x10x6 $48.90.
Motor, sensor and spindle cable	4-conductor shielded, ~45 ft	45 ft	BuildYourCNC 4-conductor shielded cable	$0.49/ft	$22.05	One spool for the steppers, switches and spindle: 18/4 shielded is enough for a 1.5-2.2 kW spindle (BuildYourCNC). Shield to earth at the VFD end. Check the gauge and that it is stranded for the cable chain.
DIN rail + terminals	UK2.5B kit	1	Erayco 20x UK2.5B + rail	$19.99	$19.99	May need more rail
Cable glands	PG7-PG16	1 (25 pcs)	Nylon cable gland kit (eBay)	$13.80	$13.80	
(opt) Tool setter	NC, magnetic base	1	Z touch tool setter	~$31.61	$31.61	Nice to have; add later

Priced subtotal: $186.19

Part 2 - Electronics: spindle (pick one)

Item	Spec	Qty	Listing	Price	Line	Notes
(choice A) 1.5 kW 110 V water-cooled kit	Spindle + 110 V VFD + 80 mm clamp + pump, ER11/ER16	1	SainSmart / VXB / Huanyang-type kits	$250-375	$300.00	Runs on a normal 120 V outlet. SainSmart $349 (sold out; open-box $249), VXB $374.88, eBay import ~$341 shipped. Check Amazon kits by hand.
(choice B) 2.2 kW 220 V kit + 120 V step-up VFD	ER20 kit + 110-120 V in / 220 V out VFD	1 + 1	RATTMMOTOR 2.2 kW kit + ATO or AT5-2200X VFD	$251 + $178-346	$500.00	Only about 1.2-1.6 kW on a 120 V circuit; full 2.2 kW once you have 240 V (then use the kit's own VFD).
(choice) ER collets	ER11/ER16 or ER20 set to match	1	Chiloskit ER20 13 pcs (or an ER11 set)	$25.21	$25.21	Skip if the kit includes a set
(choice) Spindle pump relay	SSR, DC in, AC out	1	SSR-25DA with heatsink (eBay)	$17.44	$17.44	Runs the spindle water pump whenever the spindle runs. Fuse the branch.

Item	Spec	Qty	Listing	Price	Line	Notes
(choice) Spindle coolant	Distilled water + 10-20 % RV antifreeze, bucket	1	Local store	~$10	$10.00	Water-cooled spindles only

Priced subtotal: $0.00

Part 3 - Enclosure and flood coolant

Item	Spec	Qty	Listing	Price	Line	Notes
2020 extrusion	4x766, 2x756, 4x684, 2x600, 3x560 mm	2 (10x 1000 mm)	KathyLiely 2020 T-slot 1000 mm, 10 pcs	$49.99	$99.98	One piece per stick; 5 sticks spare. Take the 2x 244 from your 400s.
ACM sheet	3 mm, all four wall/roof panels	1 (4x8 ft)	Blue Ridge Sign Supply ACM 4x8 x 3 mm, white (not Amazon)	$116.94	$116.94	Plus freight on a full sheet: get a local quote first (Piedmont Plastics, Elkridge, 301-881-7900; Central Sign Supplies lists it at $66, pickup or freight). The 726x684 sides don't fit any in-stock Amazon sheet. One 4x8 covers all four panels and the drip shield. Amazon 24x36 2-pack (B0F18ZQ7DJ, $65.99) covers back + roof only.
Door	Clear polycarbonate 560x756	1	SIBE polycarbonate 1/8 in, 24x36	$24.63	$24.63	
Corner brackets	~30	1	JCSPBYL 30 sets 2020 corner brackets	$12.99	$12.99	
T-nuts	M5 drop-in, 100	1	POLISI3D 100 M5 drop-in T-nuts	$7.65	$7.65	
Button-head screws	M5 x 10, 100	1	uxcell M5x10 button head, 100	$13.99	$13.99	
Hinges + handle	2020, 6 mm slot	1 kit	KOOTANS 2020 hinge and handle kit	$14.59	$14.59	M4 or M5 not stated
Door catch	Magnetic	1 (4-pack)	Onarway cabinet magnets	$13.99	$13.99	
(opt) Door interlock	2NC key switch	1	CZ-93B safety switch	$22.13	$22.13	Wire into the E-stop or feed-hold loop
Panel trim	U-strip for 3 mm panels	1	BLCCLOY 2020 panel U-strip, 25x 420 mm	$17.99	$17.99	10.5 m; just short of every edge
Brush seal	For 2x 72 mm motor cutouts	1 m	SLZ dust-shoe brush strip 50 mm	$20.00	$20.00	

Item	Spec	Qty	Listing	Price	Line	Notes
Bulkhead	1/4 NPT, thin panel	1 (2-pack)	FloFlex 1/4 in bulkhead, short	$9.99	$9.99	Measure the hole size first
Sump filter mesh	100 um	1	100 micron filter sock (cut open)	$10.59	$10.59	Cut open and clamp in the printed filter frame
Sump magnets	N52 bar magnets 60 x 10 x 5 mm, sealed in a zip bag	1 (12-pack)	N52 neodymium bar magnets, 12-pack	$14.99	$14.99	The bag lies on the sump floor and catches fine steel; turn it inside out to dump the chips. Was a $29.99 filter rod (B0CWGNDRNL).
Flood coolant pump	24 V brushless submersible, ~800 L/h, 5 m head, <= 60 mm tall	1	Search Amazon: "24V brushless submersible pump 800L/H 5M"	?	?	Low enough for the 4 in sump and no mains voltage in the coolant. Runs from the 24 V supply via the controller's MOSFET output. Replaces the $113 120 V pump.
Return tubing	8 mm OD PU	1 (5.5 m)	uxcell 8 mm PU tubing	$10.19	$10.19	+$1.21 shipping
Ball valve	1/4 NPT	1	Beduan brass mini ball valve	$7.99	$7.99	
Manifold	1/4 NPT Y	1	FASPARTS brass Y block	$11.99	$11.99	One bullet says 1/2 in thread: check before ordering
Loc-Line nozzles	1/4 in kit	2	Loc-Line LP-40413 coolant hose kit	$15.15	$30.30	
Push-to-connect fittings	8 mm x 1/4 NPT	1 (10)	Qjaiune push fittings	$7.49	$7.49	One at the pump outlet, so the return line unplugs and the sump slides out
Coolant concentrate	Semi-synthetic, 1 gal	1	Fusion Cool 2240	$52.95	$52.95	Mix 5-10 %. Sold out when checked on 2026-10-22: any semi-synthetic for steel will do.

Priced subtotal: $499.23 + 1 unpriced: Flood coolant pump

Already owned

- HGR15-400 rails x2 + 4 HGH15CA blocks (X)
- SFU1204-400 kit with BK10/BF10, DSG12H housing, coupler (X)
- NEMA 23 bracket for BK10 (X)
- Open-loop NEMA 23 (Z)
- Extrusion: 10x 2040 x 400, 4x 2020 x 400, 4x 2020 x 200
- Bambu A1 mini, taps, drill press

Not included

- Cutting tools, a vise and clamps (workholding).
- The water hoses that come with the spindle kit (check the barb size before buying more).
- Wire, ferrules, fuses and breakers for the electrical box.

Check before ordering

1. **Couplers.** Both ballscrew kits ship with 6.35 mm-bore couplers; your motors have 8 mm shafts. Also check your owned open-loop motor's shaft and current.
2. **VFD on 120 V.** No real 2.2 kW drive takes 120 V and outputs 220 V; the step-up drives stop at 1.5 kW. The spindle runs at up to ~1.5 kW now and gets its full 2.2 kW when you switch to the kit's 220 V VFD on 240 V. Use a dedicated 20 A circuit.
3. **No cutting.** Rails, ballscrews and tube are used at the lengths they are sold in; the risers, plates and top plates are ordered cut to size. Only the M12 threaded rod is cut (hacksaw).
4. **Tubes and rebar.** Buy plain (black) steel, not galvanized: epoxy bonds poorly to zinc. Wire-brush off rust and loose scale, degrease just before casting. Stock-length tube is +/-1/8 in on length, with an internal weld seam: file the seam where the spigots and printed sleeves go. 0.120 in wall (11 ga) works as well as 1/8 in.
5. **Imperial sizes.** The plates and risers are already inch sizes in the model. The 3/8 in top plates are 0.5 mm thinner than the modeled 10 mm, which doesn't matter.
6. **Low stock.** Several picks showed "only N left", including the closed-loop kits.